Code No: 134AK **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year II Semester Examinations, April/May-2023
COMPUTER ORGANIZATION
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Define computer architecture.	[2]
b)	Why is CPU known as the heart of the computer?	[3]
c)	What are assembly directives?	[2]
d)	List the main advantages of segmented memory scheme.	[3]
e)	What is a macro?	[2]
f)	What are the different DEBUG command characters used in MASM?	[3]
g)	What is daisy chaining interrupt?	[2]
h)	What are the three ways of representing a negative fixed-point binary numbers?	[3]
i)	What is meant by auxiliary memory?	[2]
j)	Give examples for parallel processing.	[3]

PART – B

(50 Marks)

2. With a suitable flowchart describe the concept of instruction cycle. [10]

OR

3. Demonstrate the general configuration of a micro programmed control unit. [10]

4. Describe general bus operation cycle of 8086 with suitable timing diagram. [10]

OR

5. Illustrate the functionality of the following instructions of 8086:
(a) AAM (b) LDS (c) ADC (d) XSLT (e) LODS. [10]

6. How to write an assembly language program? Explain each step in detail with a suitable program. [10]

OR

7. Write a program to find out the number of even numbers and odd numbers from a given series of 16-bit hexadecimal numbers. [10]

8. With the help of a flowchart explain the division operation on floating point numbers. [10]

OR

9.a) Make a comparison of isolated I/O with memory-mapped I/O.
b) Describe an I/O interface unit. [5+5]

10. Explain the three mapping procedures used in cache memory organization with suitable examples. [10]

OR

11. Demonstrate the usage of crossbar switch in multiprocessor systems with necessary block diagram. [10]

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